



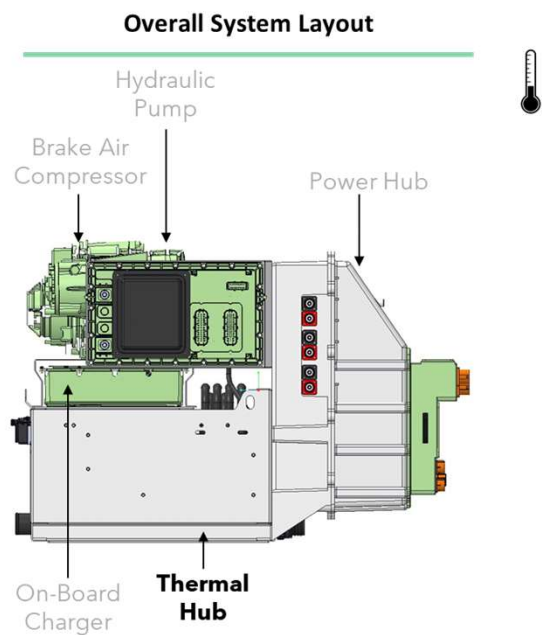
Thermal Module System (TMS) for Power Control & Accessory System

Dongmyung Suh



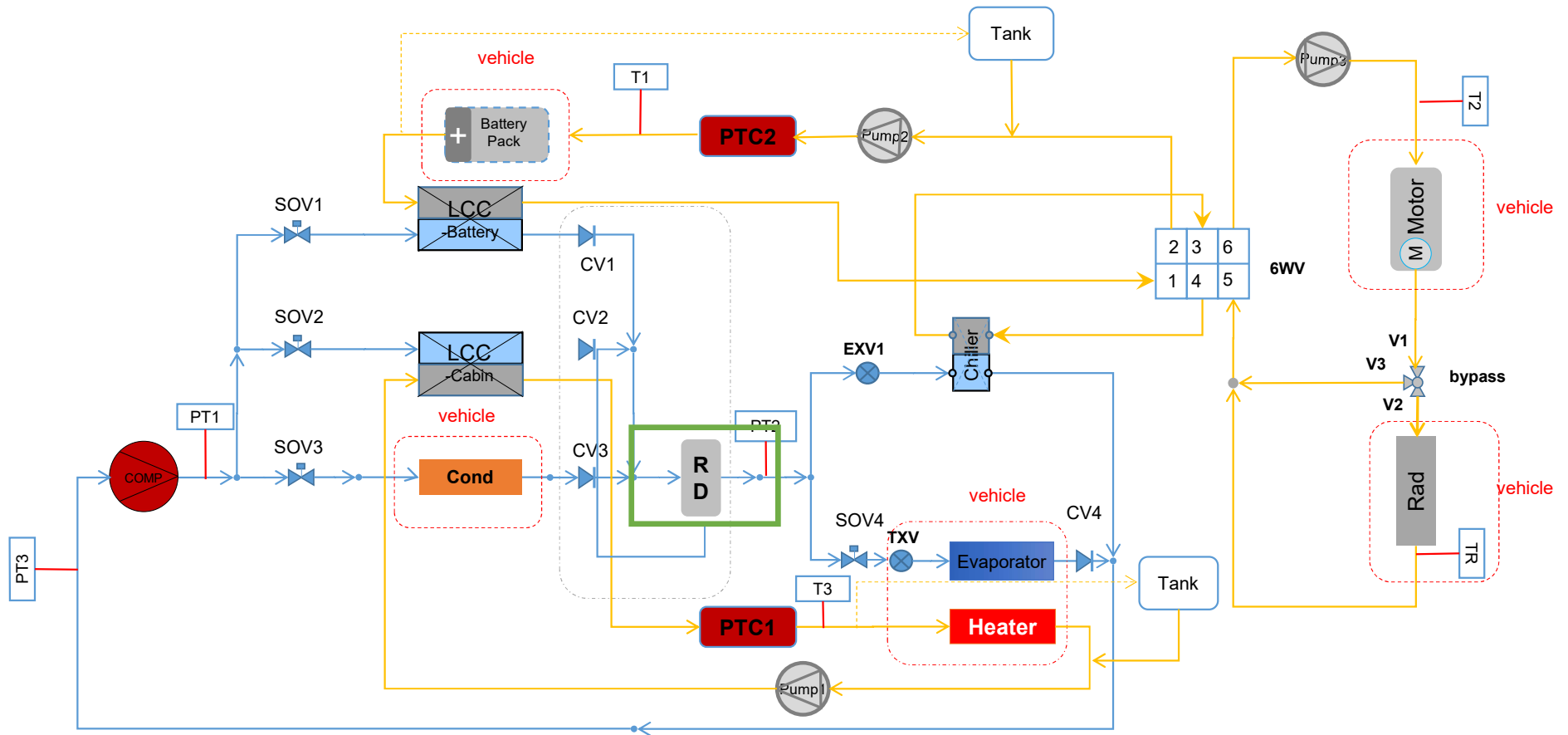
PCAS 3.0 Thermal module System Overview

System Description: TMS governs thermal-fluidic parameters for the entire vehicle, including cabin, batteries, power electronics & traction motors, heat rejection assembly (condenser, radiator, & fans), as well as the balance of plant (BOP, e.g., intercooler after intake air compressor) in the case of a fuel-cell vehicle (FCEV).



Component Performance	Unit	India Truck	MDT BEV	School Bus	HDT Day Cab BEV	HDT FCEV	HDT Sleeper Cab BEV	India Bus	Hybrid
(TMS Config)		01	02	03	04	05	06	07	08
Battery Cooling	kW	20	12	12	20	12	20	12	20
Battery Heating	kW	16	12	12	16	12	16	12	16
Battery Pump Flow Rate	lpm	160	80	80	160	80	160	80	160
Power Electronics Cooling (eAxle, PCAS within, FC BOP)	kW	21	13	13	21	44	21	13	44
Power Electronics Flow Rate	lpm	60	30	30	60	108	60	60	108
Cabin Heating	kW	0	10	22	10	10	22	0	22
Cabin Cooling	kW	7	7	-	7	7	16	0	16
Cabin Pump Flow Rate	lpm	0	20	30	20	20	30	0	30
C-Sample Cost		\$4,624	\$3,917	\$3,293	\$5,193	\$4,584	\$6,481	\$2,670	\$6,878

System Architecture



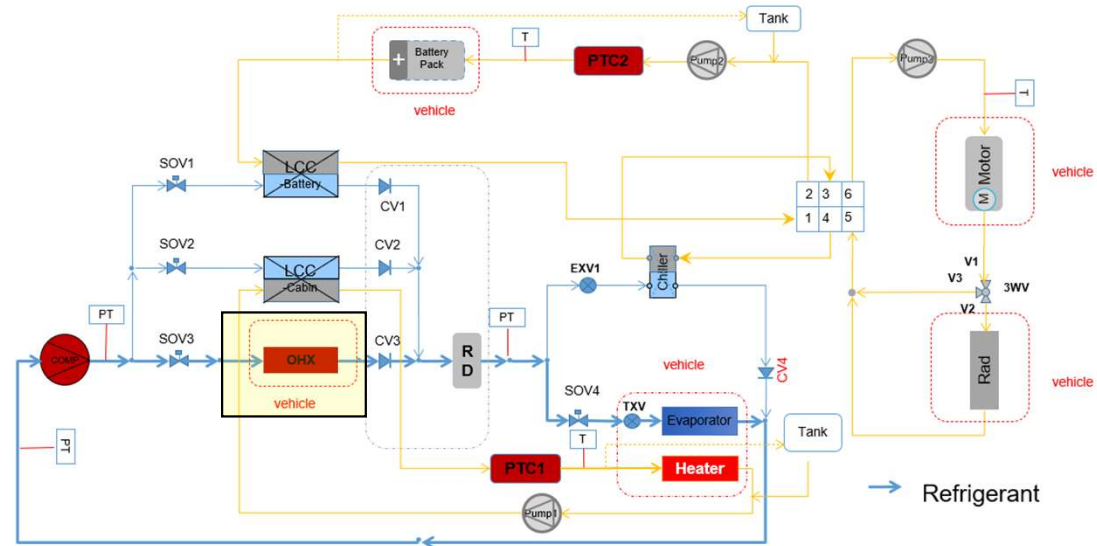
Modeling Scope

- In this modeling work, Mode 1 (Cabin Warming) and Mode 11 (Cabin Cooling) are considered.
- PCAS project currently is halted. When it resumes, the model will be extended to the other modes

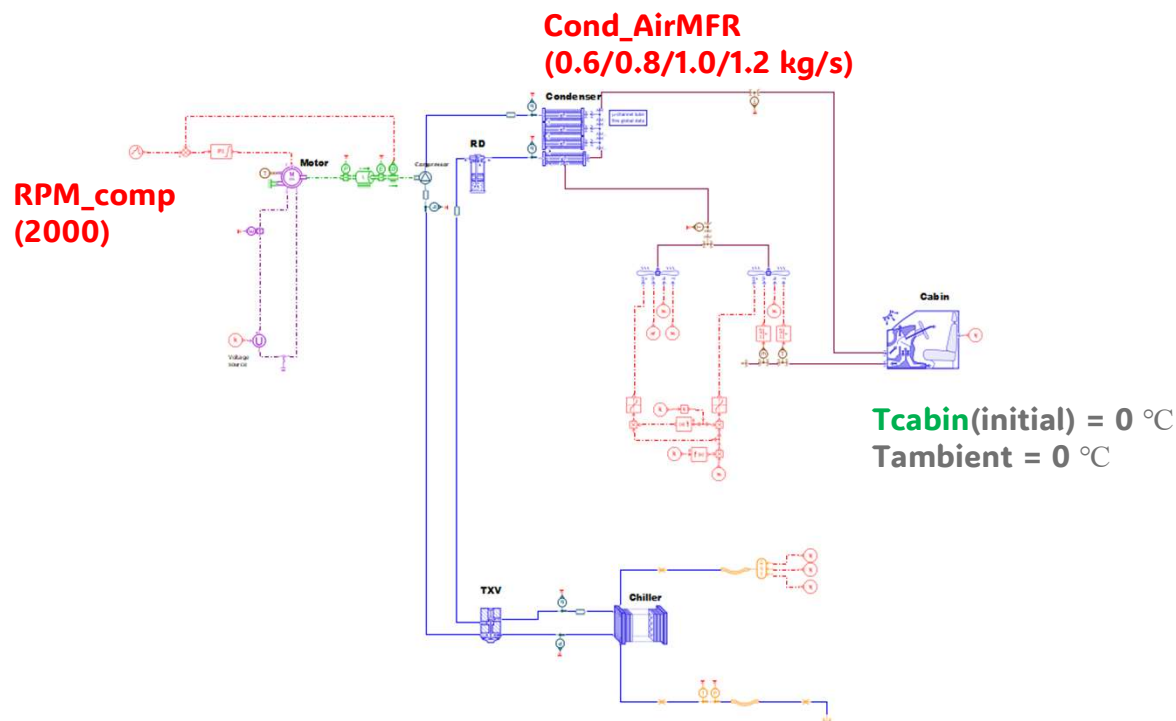
Mode 01

In this mode, TMS works for warming up the cabin. In the refrigerant cycle in the right diagram, refrigerant releases heat to air in OHX. The heated air then provides for cabin.

Cycle_Cummins			Mode	Heat Source(s)	Heat Sink(s)	Typical Ambient(°C)	Scenario										
			11	Cabin	Atmosphere	> 20°C	Driving or Cooling sleeper during overnight hot soak										
COMP	PTC1	PTC2	3WV 0%: 13pass 50%: 123all-pass 100%: 12pass default location		6WV default location	Pump1	Pump2	Pump3	SOV1	SOV2	SOV3	SOV4	EXV1	TXV	cooling fan	condensate fan	Evaporating fan
ON	OFF	OFF				OFF	OFF	OFF	OFF	OFF	ON	ON	OFF	T	OFF	ON	ON

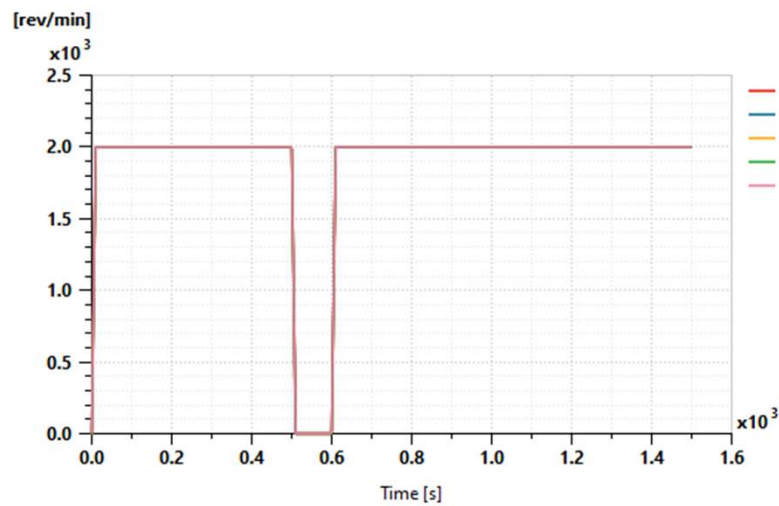


Simulation Model for Mode 01

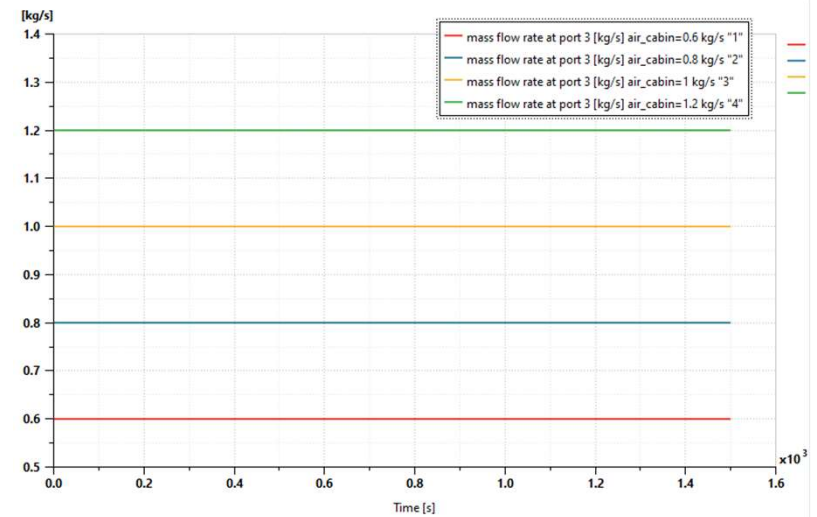


The refrigerant Cycle is modeled with Compressor powered by Motor, Condenser (OHX), RD (receiver/dryer), TXV (thermal expansion valve), and Chiller (Evaporator). Fresh air (50%) and recycled air (50%) are combined and flow through condenser. The heated air flows to cabin.





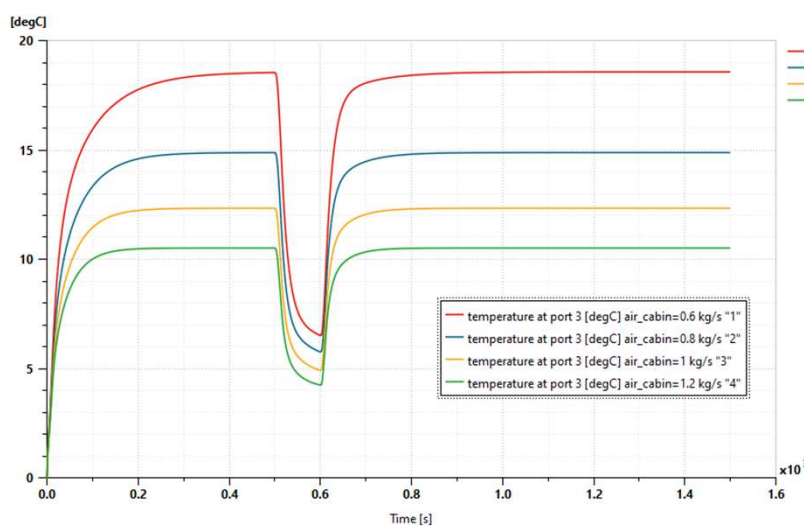
RPM_Compressor



Air MFS through Condenser

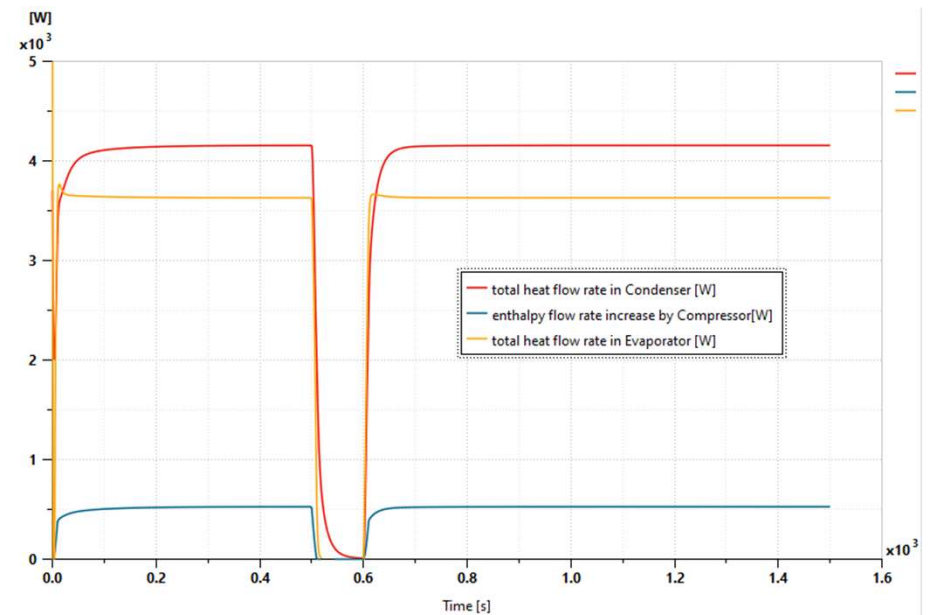
- Compressor is set to working at 2000 rpm, while it stops for 100 sec from 500 to 600 sec.
- Air flows through Condenser at constant MFS (0.6/0.8/1.0/1.2 kg/s)

- The air heated up at condenser is supplied to Cabin.
- Temperature in Cabin increases from 0 °C, initial temperature.
- As MFR of air flowing through condenser increases, its temperature increment becomes less as shown below.
- While compressor stops, cabin's temperature goes down.



Temperature in Cabin

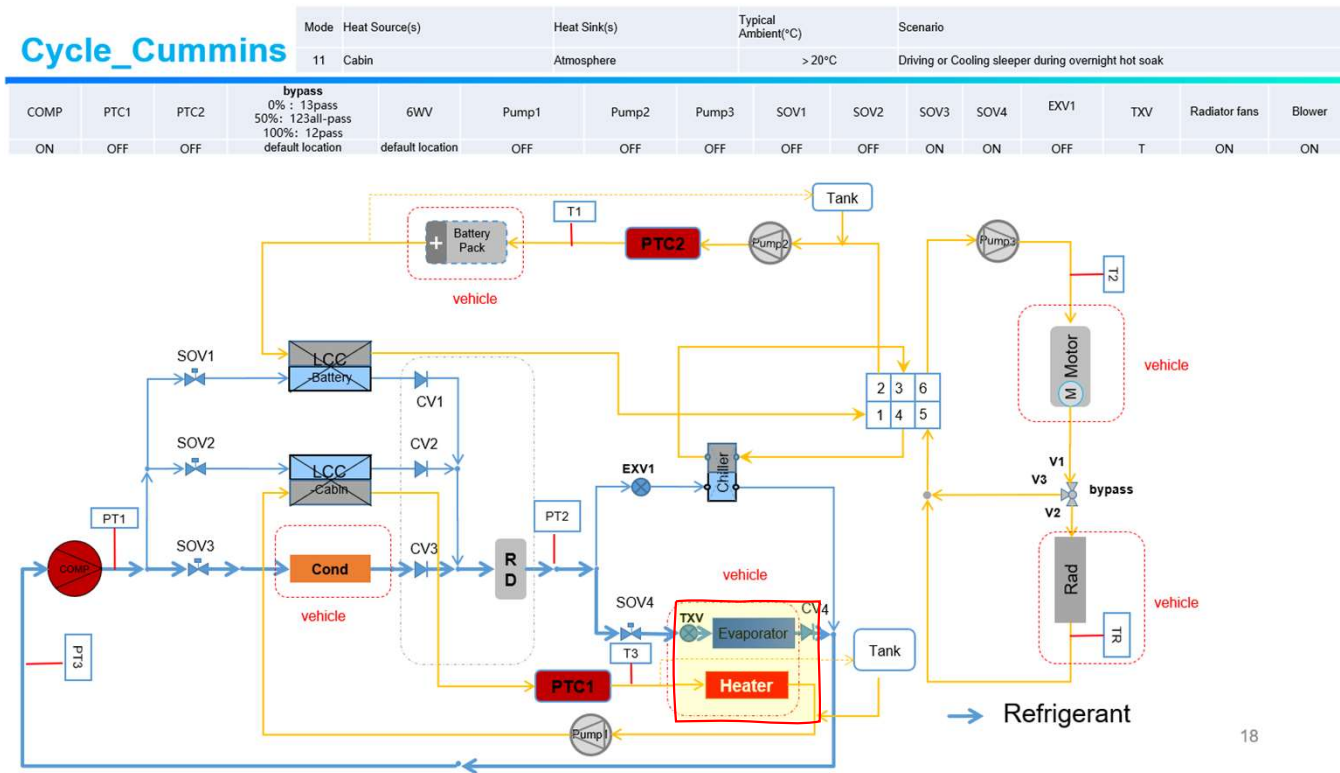
- Total heating power of condenser is shown as red curve.
- The heating power is sum of cooling power of evaporator and power provided by compressor.



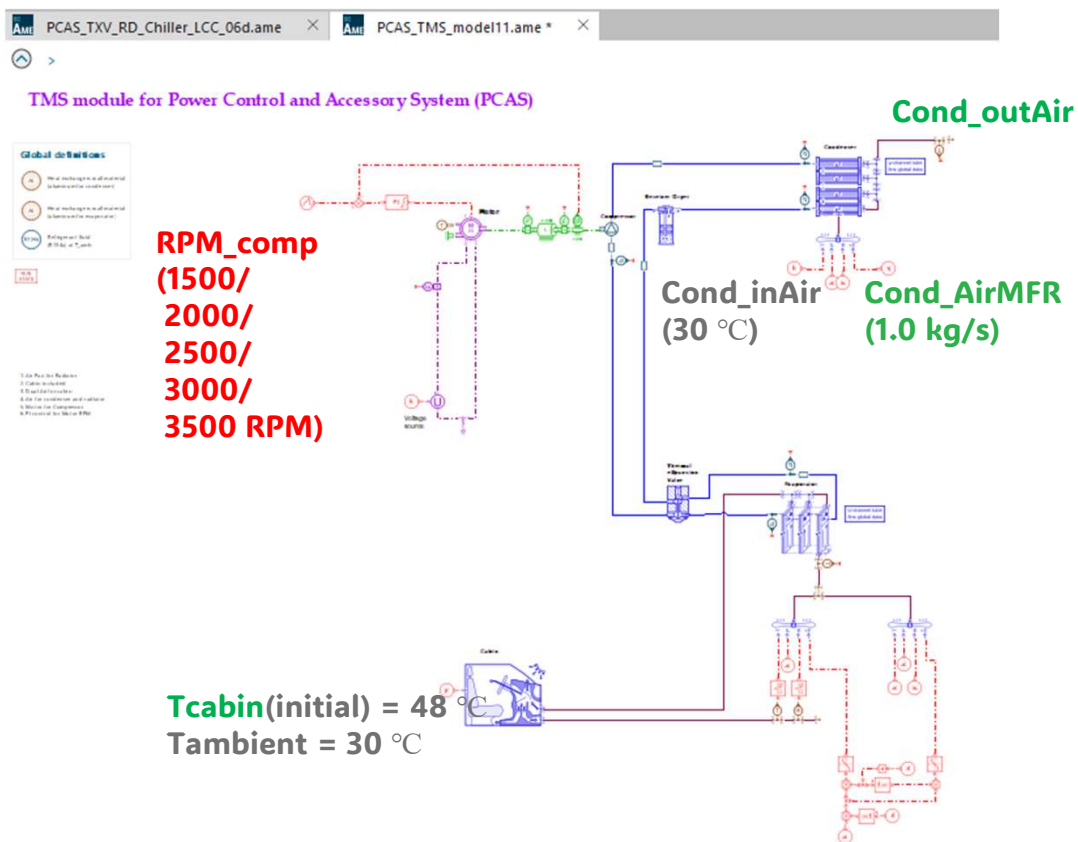
**Heating/Cooling/Power
@ air_cabin=0.6kg/s**

Mode 11

In this mode, TMS works for cooling down cabin. In Evaporator, refrigerant absorbs heat from air and the cooled air flows back to cabin.



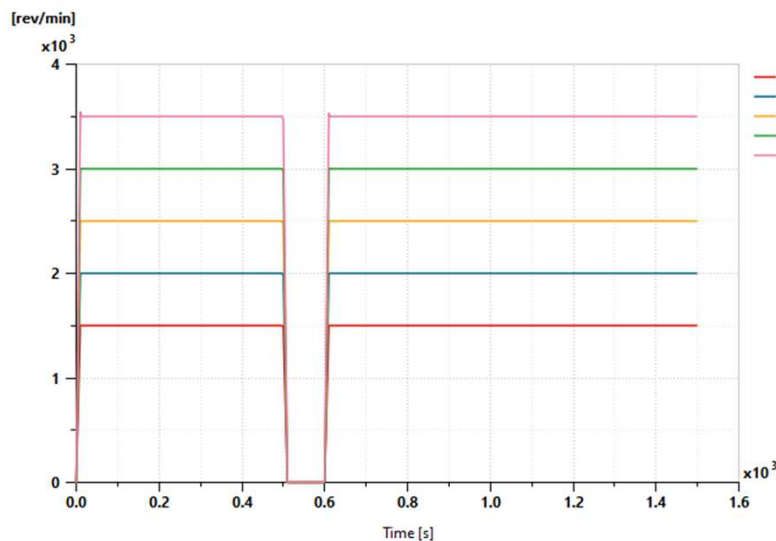
Simulation Model for Mode 11



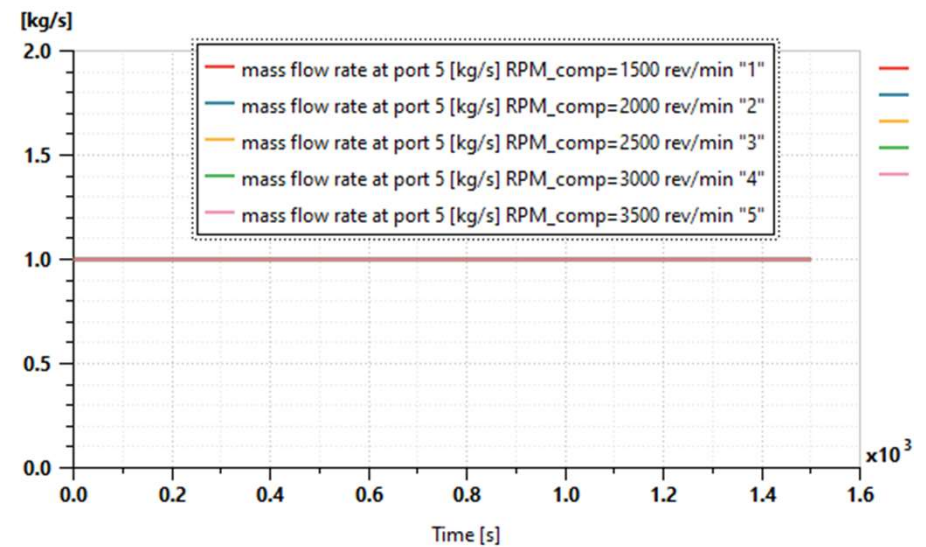
Thermal system for Mode 11 is modeled as shown in the right picture. Temperature inside cabin is 48 °C higher than ambient temperature (30 °C) due to greenhouse effect. Cabin needs to be cooled down. Cabin air flows through evaporator releasing its heat to refrigerant and then comes back to cabin at lower temperature. Compressor's RPM is set to five different values, from 1500 to 3500.

Effect of RPM_Compressor

- Compressor is set to working at five distinctive RPMs, while it stops for 100 sec from 500 to 600 sec.
- Air flows through Condenser at just 1.0 kg/s.

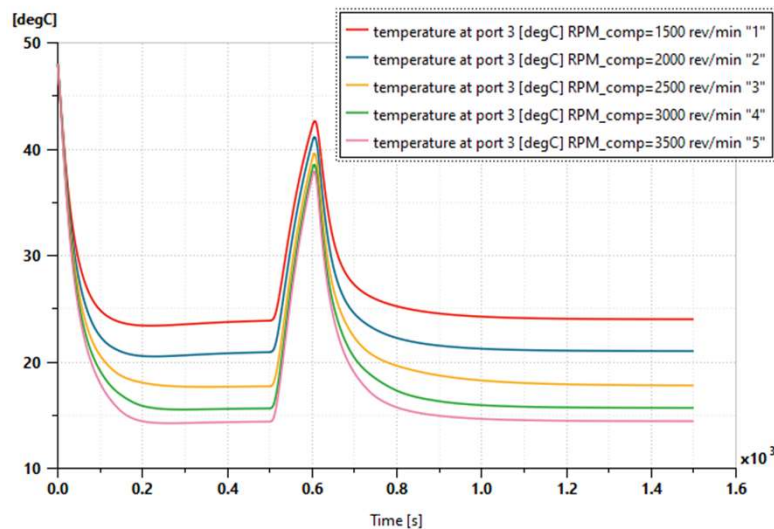


RPM_Compressor



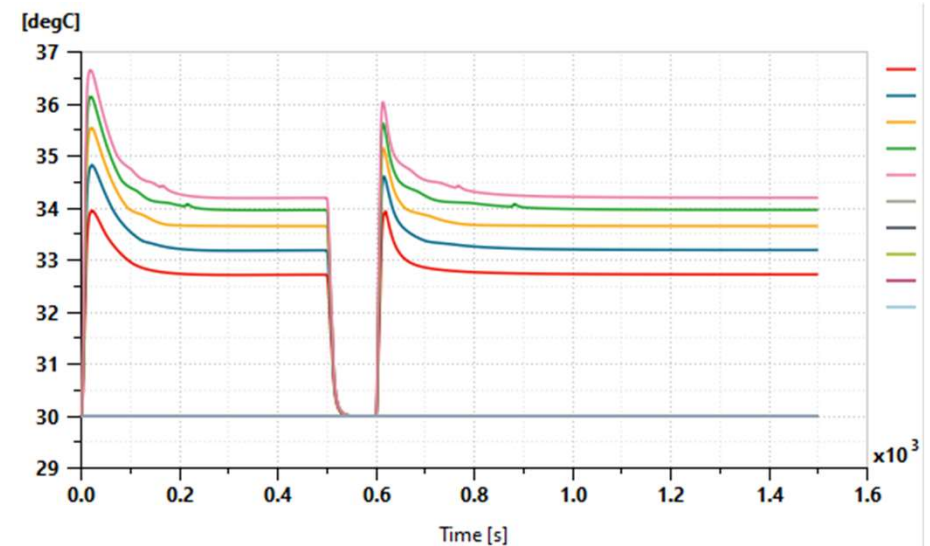
Mass Flow Rate of Air into Condenser

- When Compressor recirculates refrigerant, Air in Cabin is cooled.
- The cooled temperature is 24 °C and 14 °C at compressor RPM = 1500 and 3500 respectively.
- As RPM increases, cooling performance of evaporator goes up.



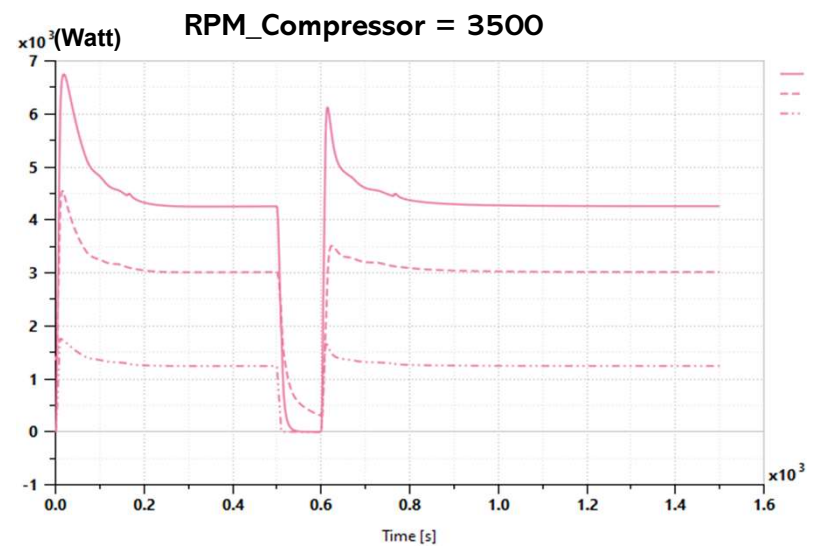
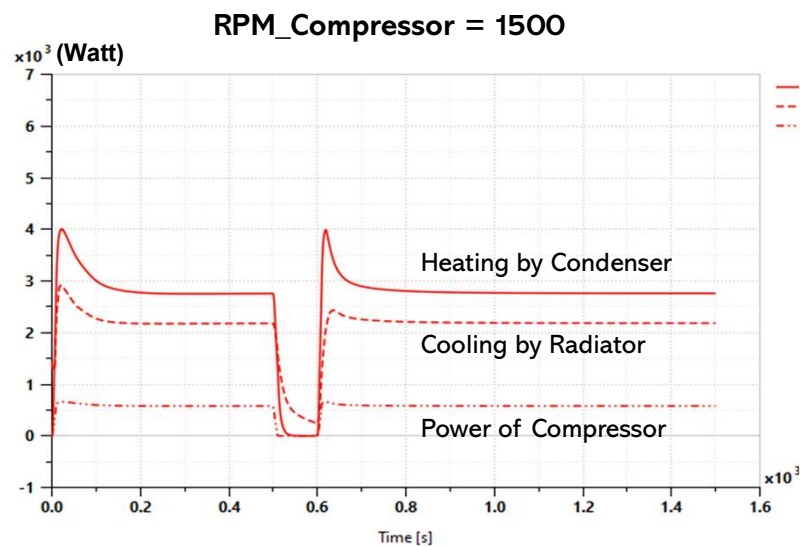
Air Temperature in Cabin

- As RPM increases, heating performance of condenser also goes up.
- Air temperature after condenser increases at RPM increases.

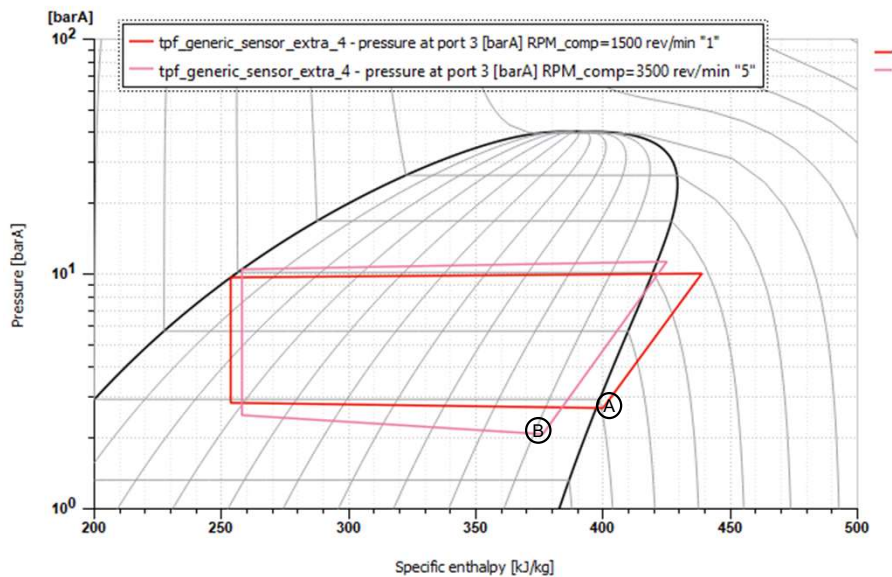


Air Temperature after Condenser

- Heating power of condenser is sum of cooling power of Radiator(evaporator) and power input to compressor.
- Magnitude of heating and cooling power are greater than the power given to compressor.
- As Compressor rotates faster (higher RPM), heating and cooling powers increases.



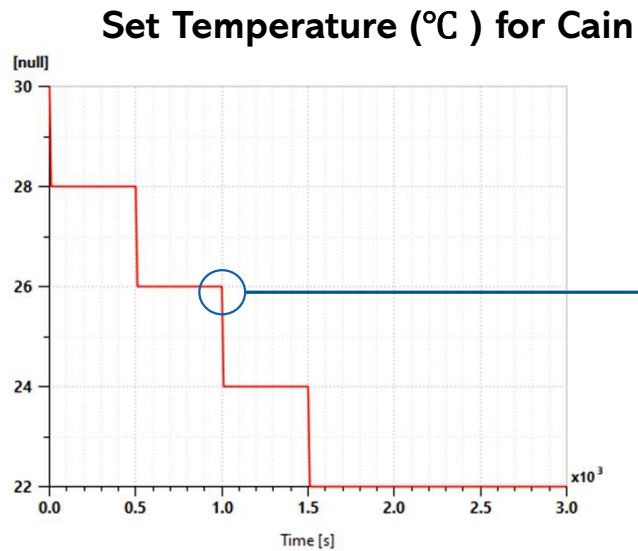
Heating/Cooling/Power



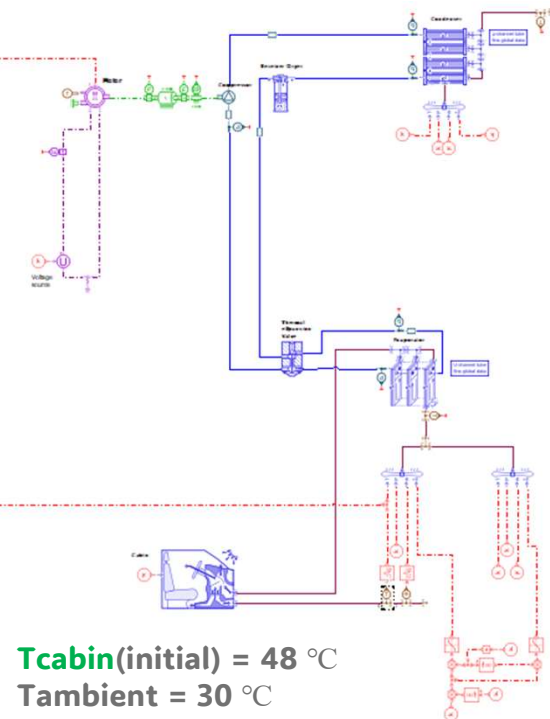
Thermodynamics

- For making refrigerant cycle work in a stable and efficient way, refrigerant should be in a gas state after evaporator. It also should be in a liquid state after condenser.
- At RPM= 1500, refrigerant is gaseous. But it is a mixture of gas and liquid at RPM=3500.
- It means that the heat absorbed by refrigerant at evaporator is not enough.
- It is advised that the area of heat exchange in evaporator should be expanded.

Control Compressor by Cabin Temp



P.I. Controller

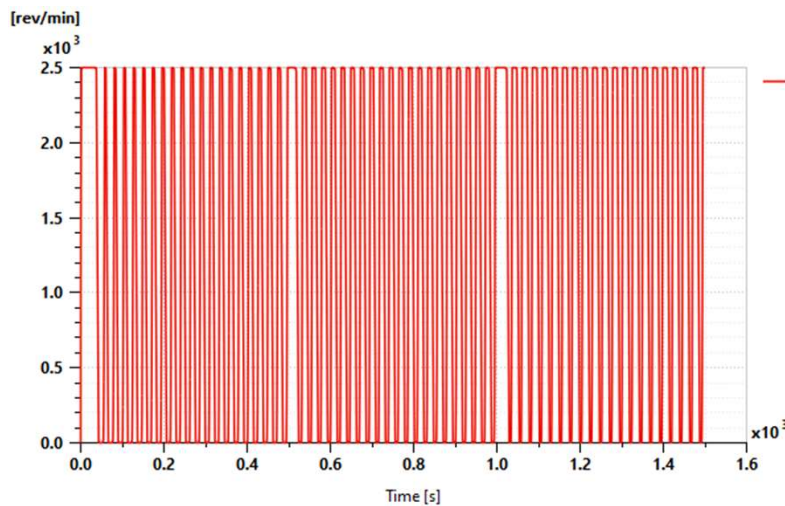


T_{cabin}(initial) = 48 °C
T_{ambient} = 30 °C

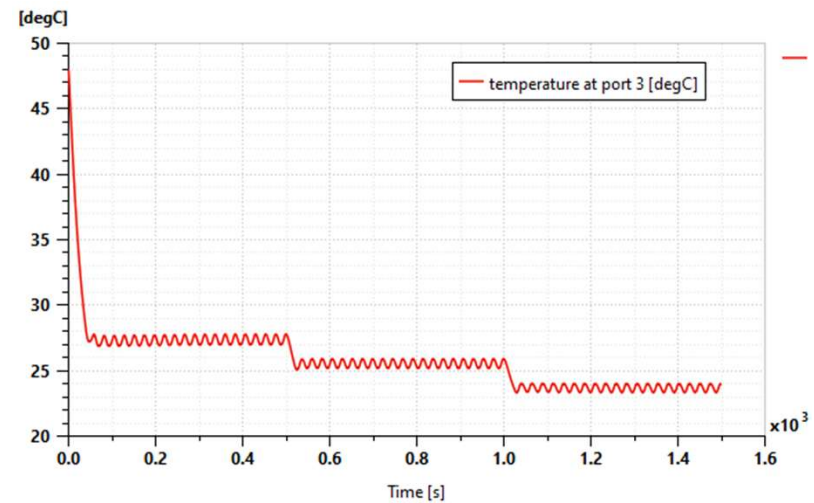
Driver sets the expected temperature for Cabin.

P.I. controller compares the setting temperature with current cabin temperature and controls torque for compressor based on the temperature difference.

- Now RPM for compressor is controlled by P.I. controller.
- The controller turns off the compressor when the cabin's temperature reaches the first setting value (28 °C)
- The temperature goes up after that. It makes the controller turn the compressor on.
- The cabin's temperature can be maintained around the setting temperature by turning the compressor on and off.
- It will occur in the same way when the cabin temperature changes to 26 and 24 °C.



RPM of Compressor



Temperature in Cabin

Conclusion

- ❑ Simulation model for Mode 1 (Cabin heating) is developed and it provides reasonable results.
- ❑ Simulation model for Mode 11 (Cabin cooling) is also developed. Using the model, the effect of Compressor rpm and thermodynamic implication are discussed.
- ❑ When PCAS project resumes, the other TMS working modes will be studied. Their simulation models are going to be developed and integrated.
- ❑ Siemens Amesim has a proper capability for modeling the complex TMS in upcoming PCAS system.